# Trøndelagkonkurranse 2026 - Level I

Third Round - October 31, 2026

2nd Part

Frida wrote the following sequence of numbers

$$011000111100000111111\ldots$$

starting with 0, then two 1's, then three 0's, then four 1's and so on. Skye then asked Frida: 'How many numbers are you going to write?'. To which she replied, 'Just 100 numbers'.

**Task 1:** Determine if the number in the 100th position is a 0 or a 1.

Skye, thrilled, started doing something similar, but instead he wrote the following sequence

$$112123123412345123456\ldots$$

each time concatenating the natural numbers from 1 to $n$ and then continuing with 1 to $n+1$. Frida then asked Skye: 'How many numbers are you going to write?'. To which he replied, 'Just 101 numbers'.

**Task 2:** Determine which the number is in the 101th position.

Peter, who came late to the meeting, saw what they were doing and also wanted to share a problem with them: either using numbers beyond the naturals (like negatives or rationals), numbers arranged in a different way other than a line, or anything interesting he can think of.

**Task 2:** Help Peter create a nice problem he can give to Frida and Skye.